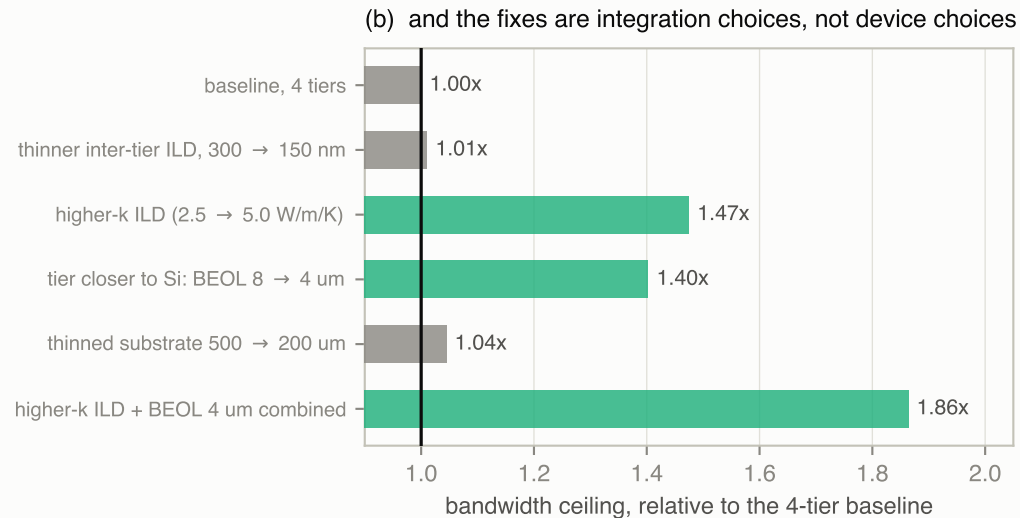
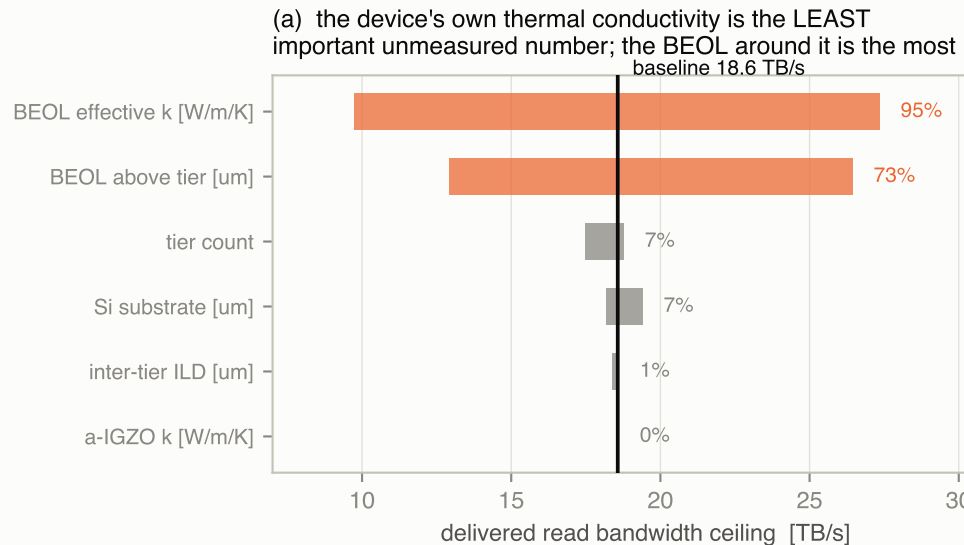


The thermal answer belongs to the BEOL integration, not to the memory device

Verified 1D transient solver (analytic x2; MAPDL 1D 0.02%; MAPDL 3D 1.3%). Ceiling is set by a temperature budget, $dT = R_{th} \times 20 \text{ W}$, so that geometry changes move R_{th} and the peak fraction independently rather than being hidden inside a fixed watt budget.



E/bit is off the left scale: 0.019-0.374 pJ/bit spans 24.8-490 TB/s, so it remains the one measurement worth asking for.

Substrate thinning is the odd one out: it lowers R_{th} but removes the thermal mass that does the duty-cycle averaging, so the two nearly cancel and only 1.04x survives. A steady-state study would have credited it with the full R_{th} improvement.